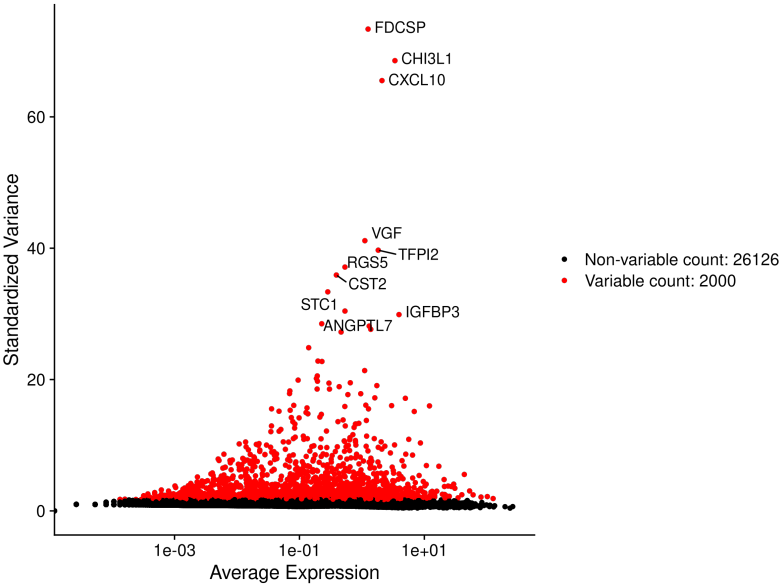


Top 10 highly variable genes - sample 1 (tumour cells)

Module 1 - Exploratory Analysis · Seurat FindVariableFeatures (vst, 2,000 features) · Summary prepared 23 September 2026



What the plot shows. After QC (500–7,500 genes per cell, <10% mitochondrial reads; 37,407 cells) and log-normalisation, 2,000 of 28,126 genes were flagged as highly variable (red). The ten labelled genes have the largest standardised variance across cells.

FDCSP, CHI3L1 and CXCL10 stand clearly apart (standardised variance ≈ 65–75), followed by a second group (VGF, TFPI2, RGS5, CST2, STC1, IGFBP3, ANGPTL7; ≈ 27–41). All sit at moderate average expression, which means they are strongly expressed in a subset of cells rather than broadly.

Gene functions

Gene	Normal function	Relevance in cancer
FDCSP Follicular dendritic cell secreted protein	Small secreted peptide made by follicular dendritic cells in lymphoid follicles; binds activated B cells and is thought to regulate B-cell function in germinal centres.	Ectopic expression reported in several carcinomas (e.g. ovarian, gastric), where it has been linked to invasion and metastasis; may also reflect tertiary lymphoid structure-like signals.
CHI3L1 Chitinase-3-like 1 (YKL-40)	Secreted glycoprotein of the chitinase family with no enzymatic activity; binds heparin, collagen and receptors such as IL-13Rα2 and RAGE to drive inflammation and tissue remodelling.	Pro-tumour: promotes angiogenesis, EMT, M2-like macrophage polarisation and T-cell suppression. Elevated serum YKL-40 is a poor-prognosis marker in many cancers; mesenchymal-state marker in glioma.
CXCL10 C-X-C motif chemokine ligand 10 (IP-10)	Interferon-γ-inducible chemokine; ligand of CXCR3. Recruits activated CD8 ⁺ and Th1 T cells and NK cells to inflamed tissue.	Readout of IFN-γ signalling in tumour cells, i.e. of active T-cell pressure. Part of 'T-cell-inflamed' signatures associated with response to immune checkpoint blockade.
VGF VGF nerve growth factor inducible	Neuropeptide precursor expressed in neurons and neuroendocrine cells; processed into bioactive peptides (e.g. TLQP-21) involved in energy balance and synaptic plasticity.	Marks neuroendocrine / neural-like tumour states; linked to stemness, EMT and resistance to targeted therapy (e.g. EGFR-TKI resistance in lung cancer, glioblastoma stem cells).
TFPI2 Tissue factor pathway inhibitor 2	Kunitz-type serine protease inhibitor (plasmin, trypsin, kallikreins); limits MMP activation and so protects the extracellular matrix.	Classically a tumour suppressor silenced by promoter methylation; in some contexts (e.g. glioblastoma) secreted TFPI2 promotes an immunosuppressive myeloid/microglia environment.
RGS5 Regulator of G-protein signalling 5	GTPase-activating protein for Gαi/Gαq that terminates GPCR signalling (e.g. endothelin, angiotensin); a canonical marker of pericytes and vascular smooth muscle.	Drives the abnormal, leaky tumour vasculature and hinders T-cell infiltration. High expression in a 'tumour cell' object may point to pericyte contamination worth checking.
CST2 Cystatin SA	Type 2 cystatin that inhibits cysteine proteases (cathepsins); normally secreted in saliva and other body fluids.	Upregulated in several carcinomas (e.g. gastric, colorectal) and associated with EMT/TGF-β signalling and poorer prognosis.

Gene	Normal function	Relevance in cancer
STC1 Stanniocalcin 1	Secreted glycoprotein hormone regulating calcium/phosphate homeostasis; strongly induced by hypoxia (HIF-1) and involved in mitochondrial metabolism.	Promotes tumour growth and angiogenesis. Described as a phagocytosis checkpoint: retains calreticulin intracellularly, reducing antigen-presenting-cell uptake and immunotherapy efficacy.
IGFBP3 Insulin-like growth factor binding protein 3	Main circulating carrier of IGF-1/IGF-2, controlling their bioavailability; also has IGF-independent pro-apoptotic and growth-inhibitory effects. Induced by p53.	Context-dependent: can be tumour-suppressive, but is also a senescence-associated secretory (SASP) factor and has been linked to therapy-resistant and mesenchymal-like tumour states.
ANGPTL7 Angiopoietin-like 7	Secreted angiopoietin-like protein highly expressed in the cornea and trabecular meshwork; regulates extracellular matrix deposition and intraocular pressure.	Hypoxia-responsive; reported to modulate tumour angiogenesis and stress responses. Less well characterised in cancer than the other genes on this list.

Take-home points

- **Immune pressure:** CXCL10 variability suggests some tumour cells are responding to IFN- γ (likely from T cells) and others are not – a useful axis for immuno-oncology questions.
- **Distinct tumour states:** CHI3L1, TFPI2, CST2 and IGFBP3 point to mesenchymal/ECM-remodelling programmes; VGF to a neuroendocrine/neural-like programme; STC1 and ANGPTL7 to hypoxia-responsive cells.
- **Possible non-tumour signal:** RGS5 (pericytes) and FDCSP (follicular dendritic cells) are classic stromal/lymphoid markers. Before clustering, it is worth checking whether a small contaminating population is driving their variance (e.g. FeaturePlot after UMAP, or co-expression with PDGFRB/ACTA2 and CR2/CLU).

Functions summarised from established literature (UniProt/NCBI Gene annotations and cancer studies); tumour-context statements are general and not specific to this dataset.